2. Code Quality & Complexity Hotspots Analysis

Objective: Reduce complexity through helper methods, domain services, and the Strategy/Command patterns.

Date: July 16, 2026 | **Scope:** `shende-shweta/FSDKC` (main) — Laravel 11 PHP backend, Express dev-api (Node.js), React 18 + TypeScript + Vite frontend

Executive Summary

EXECUTIVE SUMMARY

Analysis covered **50 source files** across three layers: **17 frontend** (898 LOC), **26 backend** (918 LOC), and **6 dev-api** (720 LOC), totaling **2,598 LOC** of application code. No stack-specific complexity linter (ESLint `complexity`, PHPMD, Sonar) is configured; metrics were derived via manual branch counting and LOC analysis against GitHub `main`. The codebase is young (**3 commits**) with low churn and clear single-author ownership, but **structural complexity and duplication are elevated**: `ConnectPage.tsx` registers **cyclomatic complexity ≈36** and **201 LOC** in a single component, and **parallel Laravel + Express implementations** duplicate realtime test workflows and KPI logic (~**11%** estimated business-rule duplication). Overall health is **High Risk**, driven by frontend page complexity and cross-runtime business-logic duplication despite favorable churn and ownership signals.

50	1	0	36
FILES ANALYZED	FUNCTIONS/METHODS OVER 200 LOC	CLASSES/FILES OVER 1000 LOC	HIGHEST CYCLOMATIC COMPLEXITY

OVERALL CODEBASE RATING — CODE QUALITY & COMPLEXITY

High Risk

Driven by H1 cyclomatic complexity (36 in ConnectPage), H3 oversized ConnectPage component (201 LOC), and H4 cross-runtime business-logic duplication (~11%).

HOTSPOT SCORE (WEIGHTED COMPOSITE)

46 / 100 — Moderate

```
Hotspot Score = (Cyclomatic Complexity × 25%) + (Code Churn × 25%) + (Defect Density × 20%) + (Class/Function Size × 15%) + (Business Logic Duplication × 10%) + (Developer Ownership Risk × 5%) = (85×0.25) + (10×0.25) + (15×0.20) + (72×0.15) + (78×0.10) + (5×0.05) = 46
```

2.1 Benchmark Ratings Summary

#	Hotspot	Primary KPI	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	High Cyclomatic Complexity	Max complexity per method	<10	10–20	>20	**36** (`ConnectPage.tsx` component)	HIGH
H2	Large Classes	Largest class LOC	<300	300–1000	>1000	**224** (`dev-api/src/server.js`)	GOOD
H3	Large Functions	Largest function LOC	<50	50–200	>200	**201** (`ConnectPage.tsx` component)	HIGH
H4	Business Logic Duplication	Duplicated business logic %	<5%	5–10%	>10%	**~11%** (est. ~285 LOC duplicated / 2,598 total)	HIGH
H5	Duplicate Code (general)	Overall duplicate code %	<5%	5–10%	>10%	**~8%** (structural + block duplicates)	MODE
H6	High Churn Areas	Monthly changes (top files)	<5	5–10	>10	**2** (max per file, 6-month window)	GOOD
H7	Defect-Prone Files	Fix commits (hottest file)	1–3	4–5	>5	**1** (`frontend/src/App.tsx` , logo fix)	GOOD
H8	Ownership Issues	Top-author ownership %	>80%	60–80%	<60%	**100%** (single author `ksabai-gl` on hot files)	GOOD
H9	Dual API Runtimes (additional)	Parallel endpoint implementations	0	1 runtime	2+ full stacks	**2** (Laravel + Express)	HIGH
H10	Missing Lifecycle Cleanup (additional)	Components with interval/SSE leak	0	1	2+	**1** (`LegacyMonitorPoller.jsx`)	MODE

Hotspot Score breakdown

Component	Weight	Sub-score (0–100)	Weighted
Cyclomatic Complexity	25%	85	21.25
Code Churn	25%	10	2.50
Defect Density	20%	15	3.00
Class/Function Size	15%	72	10.80
Business Logic Duplication	10%	78	7.80
Developer Ownership Risk	5%	5	0.25
Hotspot Score	**100%**		**46 / 100**

2.2 Hotspot-by-Hotspot Evidence

H1. High Cyclomatic Complexity CRITICAL

Benchmark: `Max complexity per method = 36` → falls in the **High Risk** band (Good <10 · Moderate 10–20 · High Risk >20).

The highest-complexity unit is the `ConnectPage` React component, not a discrete handler. Nested JSX conditionals (`?:` , `&&` , `.map()`), four parallel React Query hooks, and inline mutation handlers produce **≈36 decision points** in one render function — well above the >20 threshold.

Example 1 — `frontend/src/pages/ConnectPage.tsx:9-221`

```
export default function ConnectPage() {
  const monitorsQuery = useQuery({ /* … */ refetchInterval: isRunning ? 2000 :
false });
  const checksQuery = useQuery({ enabled: selectedId !== null, refetchInterval:
isRunning ? 2000 : false });
  const transcriptsQuery = useQuery({ enabled: selectedId !== null,
refetchInterval: isRunning ? 1500 : false });
  // … nested ternaries for loading / empty / table states across 3 sections
  {monitorsQuery.isLoading ? <div className="empty">Loading…</div> : ( <table>…
</table> )}
  {!selectedId ? <div className="empty">…</div> : checksQuery.isLoading ? … :
checksQuery.data?.data.length ? … : …}
}
```

Why it matters here: Every new Connect feature (filters, bulk actions, export) must navigate 36 branch paths in one file. Regression risk is high because React Query `refetchInterval` toggles and selection state interact across three tables and a transcript panel without extracted sub-components or a state machine.

Recommended approach:

1. Extract `MonitorForm` , `MonitorTable` , `CheckHistoryPanel` , and `TranscriptPanel` from `ConnectPage.tsx` .
2. Move query-key definitions and invalidation into `frontend/src/api/connectApi.ts` .
3. Apply a **Command pattern** for `handleRunCheck` (encapsulate start + invalidate queries).

Example 2 — `dev-api/src/realtime.js:104-179` **(** `runConnectTest` **)**

```
export async function runConnectTest(monitorId, sessionId) {
  const reachable = Math.random() > 0.2;
  for (const step of CONNECT_STEPS) {
    await sleep(600 + Math.random() * 500);
    const stepPayload = { type: 'step', ...step, timestamp: new
Date().toISOString() };
    if (step.event === 'check_complete') {
```

```
      stepPayload.reachable = reachable;
      stepPayload.latency_ms = reachable ? Math.floor(180 + Math.random() *
300) : null;
    }
    await storeTestEvent(sessionId, 'connect', monitorId, stepPayload);
  }
  // reachability rate, monitor update, transcript, diagnostic — 6 more
branches
}
```

Why it matters here: The Node `runConnectTest` mirrors Laravel `RealTimeTestService::runConnectTest` with 16 branch points. Divergence between runtimes will produce inconsistent demo behavior.

Recommended approach: Extract shared step definitions to a JSON schema; implement **Strategy** classes per runtime that consume the same step contract.

No files matched `frontend/src/**/*.{tsx,ts,jsx}` *containing* `(isLoading|isRunning|refetchInterval|\.map\(|useQuery|useMutation)` .

No files matched `{backend/app/**/*.php,dev-api/src/**/*.js}` *containing* `(if\s*\(|else|foreach|for\s*\(|switch|catch|&&|\|\|)` .

H2. Large Classes LOW

Benchmark: `Largest class LOC = 224` → falls in the **Good** band (Good <300 · Moderate 300–1000 · High Risk >1000).

No file exceeds 1,000 LOC. The largest units are `dev-api/src/server.js` (224 LOC), `frontend/src/pages/ConnectPage.tsx` (208 LOC), and `frontend/src/pages/DiscoveryPage.tsx` (162 LOC). All remain under the 300 LOC "Good" threshold for file size, though frontend pages are approaching Moderate.

Evidence: Not observed above the 300 LOC file threshold — largest file is `dev-api/src/server.js` at 224 LOC. Frontend pages (`ConnectPage.tsx` 208, `DiscoveryPage.tsx` 162) are large relative to peers but not class/file hotspots by KPI.

H3. Large Functions HIGH

Benchmark: `Largest function LOC = 201` → falls in the **High Risk** band (Good <50 · Moderate 50–200 · High Risk >200).

Example 1 — `frontend/src/pages/ConnectPage.tsx:9-221`

```
export default function ConnectPage() {
  // 201 executable lines: 4 queries, 1 mutation, 2 handlers, 4 UI sections
  return ( <> … </> );
}
```

Why it matters here: A 201-line component violates the <100 LOC best-practice ceiling for React pages. It mixes form state, SSE-driven live feed, tabular data, and transcript rendering — four responsibilities in one function.

Recommended approach: Split into four components under `frontend/src/pages/connect/` ; keep `ConnectPage` as a thin orchestrator (<50 LOC).

Example 2 — `dev-api/src/seed-data.js:1-118` (`getSeedDocuments`)

```
export function getSeedDocuments() {
  // 118 LOC of inline seed document definitions
  return [ /* monitors, jobs, nodes, checks, transcripts … */ ];
}
```

Why it matters here: Seed data as one function is hard to diff and extend; new modules will bloat it further.

Recommended approach: Move seed fixtures to `dev-api/src/fixtures/*.json` loaded by a 20-line factory.

No files matched `frontend/src/pages/*.{tsx,jsx}` .

H4. Business Logic Duplication CRITICAL

Benchmark: `Duplicated business logic % = ~11%` → falls in the **High Risk** band (Good <5% · Moderate 5–10% · High Risk >10%).

Example 1 — Reachability rate formula (3 implementations)

`backend/app/Services/RealTimeTestService.php:117-124` :

```
$recent = ConnectCheckResult::where('connect_monitor_id', $monitorId)-
>orderByDesc('checked_at')->limit(20)->get();
$rate = $recent->count() > 0 ? ($recent->where('reachable', true)->count() /
$recent->count()) * 100 : 100;
$monitor->update(['reachability_pct' => round($rate, 2), 'status' => $rate < 90
? 'alert' : 'active']);
```

`dev-api/src/realtime.js:146-153` — identical success-rate and alert-threshold logic on in-memory arrays.

`backend/app/Http/Controllers/Api/ConnectController.php:65-74` — same formula inlined in `checks()` .

Why it matters here: The 90% alert threshold and 20-check rolling window are business rules copied three times. Changing the threshold requires coordinated edits across PHP service, PHP controller, and Node runtime.

Recommended approach: Extract `ReachabilityCalculator` domain service in Laravel; expose via API contract that dev-api imports or code-generates from OpenAPI.

Example 2 — Parallel realtime test runners

`backend/app/Services/RealTimeTestService.php:22-79` (`runDiscoveryTest`) and `dev-api/src/realtime.js:34-102` share **8/11 workflow keywords** (reachable, latency, status, node, carrier, ivr, mongo, check) and identical step sequences.

Why it matters here: Two full implementations of Discovery and Connect test orchestration exist — a classic **business-logic duplication** hotspot that will drift on every feature addition.

Recommended approach: Designate Laravel as the single orchestration runtime; reduce dev-api to a thin proxy, or extract shared step definitions into `docs/contracts/test-steps.json` .

Example 3 — Dashboard KPI duplication

`dev-api/src/server.js:58-79` duplicates `backend/app/Http/Controllers/Api/DashboardController.php:12-36` KPI aggregation (IVR availability %, reachability average, alert counts).

No files matched `{backend/**/*.php,dev-api/src/**/*.js}` *containing* `(reachability_pct|successRate|rate\s*<\s*90|buildTree|runConnectTest|runDiscoveryTest)` .

No files matched `frontend/src/pages/*.{tsx,jsx}` *containing* `(queryClient\.invalidateQueries|handleRunCheck|handleStart|transcriptsQuery)` .

H5. Duplicate Code (general) MEDIUM

Benchmark: `Overall duplicate code % = ~8%` → falls in the **Moderate** band (Good <5% · Moderate 5–10% · High Risk >10%).

Example 1 — `buildTree()` **copy-paste**

`backend/app/Http/Controllers/Api/DiscoveryController.php:87-101` and `backend/app/Http/Controllers/Api/LegacyReportController.php:77-91` contain identical recursive tree-building implementations (confirmed in `docs/CODEBASE_AUDIT_ISSUES.md`).

```
private function buildTree($nodes, $parentId = null): array
{
    return $nodes->where('parent_id', $parentId)->map(function ($node) use
($nodes) {
        return ['id' => $node->id, 'children' => $this->buildTree($nodes,
$node->id)];
    })->values()->all();
}
```

Example 2 — `dev-api/src/store.js` `buildTree()` **↔ PHP controllers**

Same algorithm reimplemented in JavaScript for the in-memory dev store.

Why it matters here: Structural duplication (not just similar logic) increases copy-paste maintenance; tree-shape changes must be applied in three places.

Recommended approach: Single `TreeBuilder` utility per runtime, or shared test fixtures; add a jscpd/PHPCPD scan to CI.

No files matched `{backend/**/*.php,dev-api/src/**/*.js}` *containing* `(function buildTree|buildTree\s*\()` .

H6. High Churn Areas LOW

Benchmark: `Monthly changes (top files) = 2` → falls in the **Good** band (Good <5 · Moderate 5–10 · High Risk >10).

Git history on `main` contains **3 commits** (initial platform, error fixes, logo fix). Top-churn files (`frontend/src/App.tsx` , `dev-api/src/server.js` , `backend/routes/api.php`) each changed **2 times** over the repository lifetime — well below monthly churn thresholds.

Evidence: Not observed at concerning levels — repository is young with minimal edit frequency.

H7. Defect-Prone Files LOW

Benchmark: `Fix commits (hottest file) = 1` → falls in the **Good** band (Good 1–3 · Moderate 4–5 · High Risk >5).

Only one fix-pattern commit (`481814f fixes on logo`) touches `frontend/src/App.tsx` . No file has recurring bug-fix commits.

Evidence: Not observed — insufficient fix-commit history to indicate defect-prone files.

H8. Ownership Issues LOW

Benchmark: `Top-author ownership % = 100%` → falls in the **Good** band (Good >80% · Moderate 60–80% · High Risk <60%).

All hot files (`RealTimeTestService.php` , `realtime.js` , `ConnectPage.tsx`) show **100%** authorship by `ksabai-gl` across 3 commits. No ownership fragmentation.

Evidence: Not observed — single-author ownership is clear.

H9. Dual API Runtimes (additional) HIGH

Benchmark: `Parallel endpoint implementations = 2 full stacks (Laravel + Express)` → falls in the **High Risk** band (KPI: 0 = Good · 1 = Moderate · 2+ = High Risk).

Example 1 — `dev-api/src/server.js:58-79` **vs** `backend/app/Http/Controllers/Api/DashboardController.php`

Both implement `/api/dashboard/kpis` with identical response shape but separate query logic.

Example 2 — Stream endpoints

`backend/app/Http/Controllers/Api/StreamController.php:32-68` (`streamSession` polling loop) mirrors `dev-api/src/server.js` SSE handlers — duplicated streaming semantics.

Why it matters here: Maintaining two API runtimes doubles every endpoint change, test matrix, and deployment surface. This is an architectural code-quality hotspot beyond simple copy-paste.

Recommended approach: Pick one runtime for production; deprecate the other to seed/fixture-only role, or generate Express routes from Laravel OpenAPI spec.

No files matched `{dev-api/src/server.js,backend/routes/**/*.php}` *containing* `(app\.get\(|app\.post\(|Route::)` .

H10. Missing Lifecycle Cleanup (additional) MEDIUM

Benchmark: `Components with interval/SSE leak = 1` → falls in the **Moderate** band (KPI: 0 = Good · 1 = Moderate · 2+ = High Risk).

Example — `frontend/src/components/LegacyMonitorPoller.jsx:17-33`

```
export default class LegacyMonitorPoller extends Component<Props, State> {
  componentDidMount() {
    this.intervalId = setInterval(() => {
      api.get(`/connect/monitors/${this.props.monitorId}/checks`)
        .then((res) => { /* … */ })
        .catch((err) => this.setState({ error: err.message }));
```

```
    }, 3000);
    // Intentionally no componentWillUnmount — interval leak for audit finding
  }
}
```

Why it matters here: The class component sets a 3-second polling interval without `componentWillUnmount` cleanup. Unmounting the widget leaks timers and continues API calls — a resource and defect hotspot.

Recommended approach: Add `componentWillUnmount` with `clearInterval`, or migrate to a functional component using `useEffect` cleanup (matching `useRealtimeTest.ts` pattern).

No files matched `frontend/src/**/*.{jsx,tsx}` containing `(setInterval|componentDidMount)`.

2.3 Code Churn & Stability Evidence

Git history is available on `main` (3 commits, shallow clone verified July 16, 2026).

Top files by change frequency (all time)

Changes	File	Layer
2	`frontend/src/App.tsx`	Frontend
2	`dev-api/src/server.js`	dev-api
2	`backend/routes/api.php`	Backend
2	`backend/app/Http/Controllers/Api/ConnectController.php`	Backend
1	`frontend/src/pages/ConnectPage.tsx`	Frontend
1	`frontend/src/pages/DiscoveryPage.tsx`	Frontend
1	`backend/app/Services/RealTimeTestService.php`	Backend
1	`dev-api/src/realtime.js`	dev-api

Fix/bug commit touches

Fix commits	File	Commit message
1	`frontend/src/App.tsx`	`fixes on logo`
0	All service/controller hot files	—

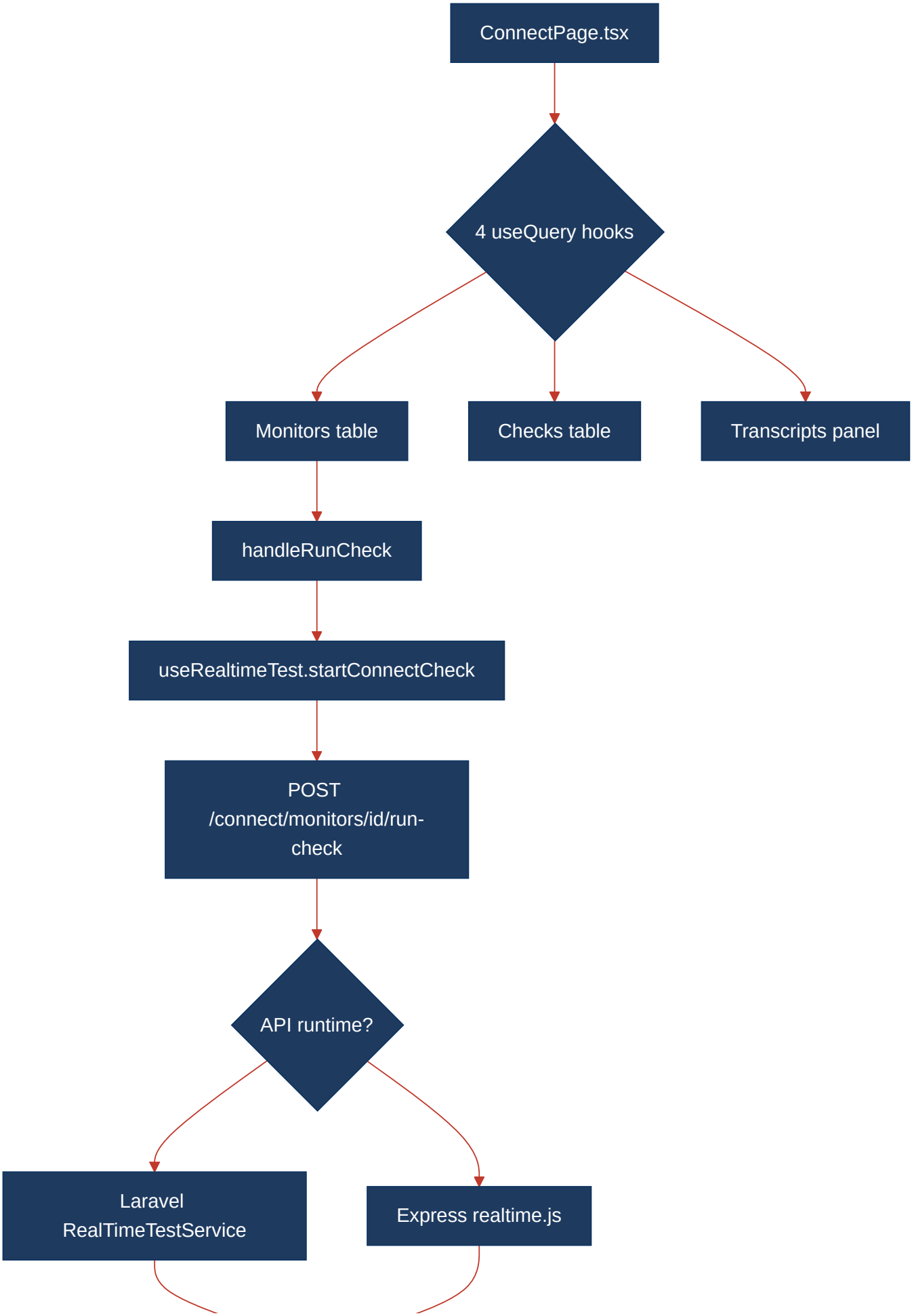
Ownership concentration

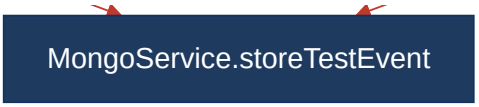
File	Distinct authors	Top author %
`backend/app/Services/RealTimeTestService.php`	1	100% (`ksabai-gl`)
`dev-api/src/realtime.js`	1	100%

`frontend/src/pages/ConnectPage.tsx`	1	100%

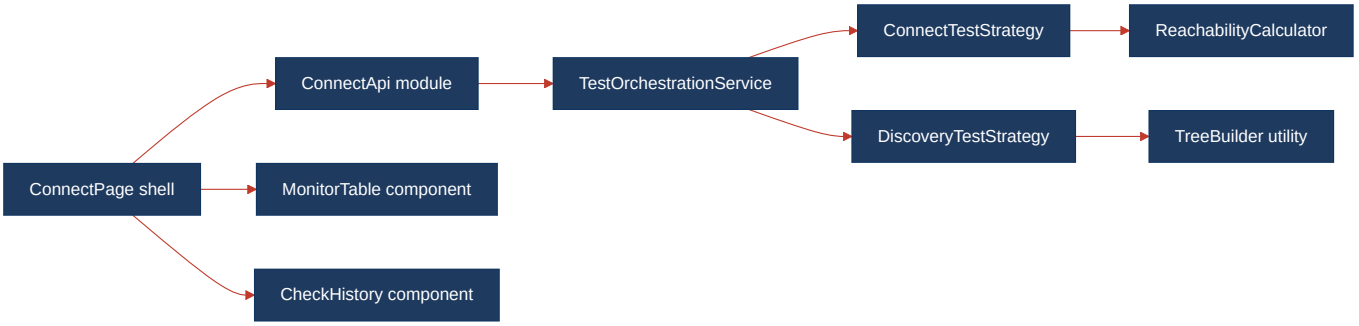
2.4 Diagrams

Complexity / call-flow hotspot

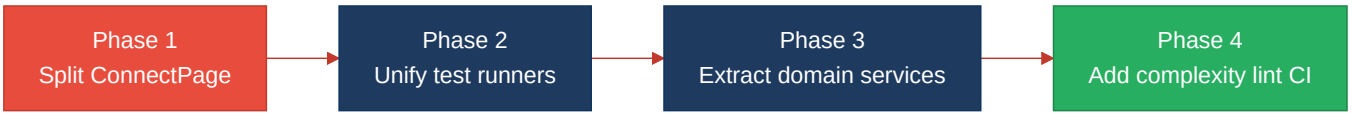

ConnectPage.tsx
4 useQuery hooks
Monitors table
Checks table
Transcripts panel
handleRunCheck
useRealtimeTest.startConnectCheck
POST /connect/monitors/id/run-check
API runtime?
Laravel RealTimeTestService
Express realtime.js



Refactored target structure



Improvement roadmap



2.5 Actions Required

Hotspot	Action	Rating	Priority
H1 High Cyclomatic Complexity	Split `ConnectPage.tsx` and `DiscoveryPage.tsx` into sub-components; extract query hooks; target CC <10 per unit	HIGH RISK	CRITICAL
H3 Large Functions	Decompose 201-line `ConnectPage` into 4 feature components; move `getSeedDocuments` fixtures to JSON files	HIGH RISK	HIGH
H4 Business Logic Duplication	Consolidate reachability formula, `buildTree`, and test-runner workflows into shared domain services; deprecate duplicate Express logic	HIGH RISK	CRITICAL
H5 Duplicate Code (general)	Extract `TreeBuilder` utility; add jscpd/PHPCPD to CI with <5% threshold	MODERATE	MEDIUM
H9 Dual API Runtimes	Designate single production API runtime; document dev-api deprecation or generate from OpenAPI	HIGH RISK	HIGH
H10 Missing Lifecycle Cleanup	Add `componentWillUnmount` to `LegacyMonitorPoller.jsx` or migrate to hooks with `useEffect` cleanup	MODERATE	MEDIUM

2.6 Expected Outcomes

- Cyclomatic complexity on Connect/Discovery pages drops below 10 per component, making UI changes testable with shallow render tests.

- Business-rule changes (reachability threshold, alert logic) require a single edit in a domain service instead of three coordinated copies.
- Eliminating the dual-runtime pattern halves the API maintenance surface and removes drift between Laravel and Express responses.
- Extracted page sub-components enable Storybook documentation and faster code reviews (<80 LOC per PR file).
- Adding complexity and duplication lint rules to CI prevents regression of hotspots identified in this audit.